in listing 10.10. This exercise example performs better than the previous example on the same problem.

Open the `cot.jinja2` prompt in VS Code, as shown in listing 10.11. This is a much simpler prompt than the previous example because it only uses zero-shot. However, one key phrase turns this simple prompt into a powerful reasoning engine. The line in the prompt `Let's think step by step` triggers the LLM to consider internal context showing reasoning. This, in turn, directs the LLM to reason out the problem in steps.

Listing 10.11 `cot.jinja2`

```
system:
You are an expert in solving time travel problems.
You are given a time travel problem and you have to solve it.
Let's think step by step.     #1
Please finalize your answer in a single statement.     #2

user:
{{statement}}     #3
```

#1 A magic line that formulates reasoning from the LLM
#2 Asks the LLM to provide a final statement of the answer
#3 The problem statement the LLM is asked to solve

Similar phrases asking the LLM to think about the steps or asking it to respond in steps also extract reasoning. We'll demonstrate a similar but more elaborate technique in the next section.

10.2.3 Step by step with prompt chaining

We can extend the behavior of asking an LLM to think step by step into a chain of prompts that force the LLM to solve the problem in steps. In this section, we look at a technique called *prompt chaining* that forces an LLM to process problems in steps.

Open the `prompt_flow/prompt-chaining/flow.dag.yaml` file in the visual editor, as shown in figure 10.5. Prompt chaining breaks up the reasoning method used to solve a problem into chains of prompts. This technique forces the LLM to answer the problem in terms of steps.